

Nupur Kumari

Graduate Student
Robotics Institute
Carnegie Mellon University

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Education

Carnegie Mellon University

Ph.D. in Robotics, Advisor: Jun-Yan Zhu 2022 - 2026
Thesis: Customizing Generative Image Models
Thesis Committee: Jun-Yan Zhu (CMU)
Deva Ramanan (CMU)
Shubham Tulsiani (CMU)
Phillip Isola (MIT)

MS in Robotics, Advisor: Jun-Yan Zhu, (GPA: 4.17/4.3) 2021 - 2022

Indian Institute of Technology Delhi

Integrated M. Tech in Mathematics and Computing (GPA: 9.15/10.0) 2012 - 2017

Work Experience

Meta (GenAI), US

Summer Research Intern

2024

Adobe Research, US

Summer Research Intern

2022, 2023, 2025

Adobe, India

MTS (Media and Data Science Research Lab)

2017-2020

Selected Publications [\[Google Scholar\]](#)

- **Nupur Kumari**, Sheng-Yu Wang, Nanxuan Zhao, Yotam Nitzan, Yuheng Li, Krishna Kumar Singh, Richard Zhang, Eli Shechtman, Jun-Yan Zhu, Xun Huang. *NP-Edit: Learning an Image Editing Model without Image Editing Pairs*. ICLR 2026. [\[Paper\]](#).
- **Nupur Kumari**, Xi Yin, Jun-Yan Zhu, Ishan Misra, Samaneh Azadi. *Generating Multi-Image Synthetic Data for Text-to-Image Customization*. ICCV 2025. [\[Paper\]](#).
- Sean Liu, **Nupur Kumari**, Ariel Shamir, Jun-Yan Zhu. *Generative Photomontage*. CVPR 2025. [\[Paper\]](#).
- **Nupur Kumari**, Grace Su, Richard Zhang, Taesung Park, Eli Shechtman, Jun-Yan Zhu. *Customizing Text-to-Image Diffusion with Camera Viewpoint Control*. SIGGRAPH Asia 2024. [\[Paper\]](#).
- Maxwell Jones, Sheng-Yu Wang, **Nupur Kumari**, David Bau, Jun-Yan Zhu. *Customizing Text-to-Image Models with a Single Image Pair*. SIGGRAPH Asia 2024. [\[Paper\]](#).
- **Nupur Kumari**, Bingliang Zhang, Sheng-Yu Wang, Eli Shechtman, Richard Zhang, Jun-Yan Zhu. *Ablating Concepts in Text-to-Image Diffusion Models*. ICCV 2023. [\[Paper\]](#).
- **Nupur Kumari**, Bingliang Zhang, Richard Zhang, Eli Shechtman, Jun-Yan Zhu. *Multi-Concept Customization of Text-to-Image Diffusion*. CVPR 2023. [\[Paper\]](#). **The top-10 cited papers among all CVPR-2023 papers, refer to here.**
- Daohan Lu*, Sheng-Yu Wang*, **Nupur Kumari***, Rohan Agarwal*, Mia Tang, David Bau, Jun-Yan Zhu. *Content-Based Search for Deep Generative Models*. SIGGRAPH Asia 2023. [\[Paper\]](#).
- **Nupur Kumari**, Richard Zhang, Eli Shechtman, Jun-Yan Zhu. *Ensembling Off-the-shelf Models for GAN Training*. CVPR 2022 (Oral). [\[Paper\]](#). **Selected for oral presentation at the conference. Selection rate: 4.2% (344 papers selected out of 8161 submissions)**
- Mayank Singh*, **Nupur Kumari***, Puneet Mangla, Abhishek Sinha, Balaji Krishnamurthy, Vineeth N Balasubramanian. *Attributional Robustness Training using Input-Gradient Spatial Alignment*. ECCV 2020. [\[Paper\]](#).
- **Nupur Kumari***, Mayank Singh*, Abhishek Sinha*, Harshitha Machiraju, Balaji Krishnamurthy, Vineeth N Balasubramanian. *Harnessing the Vulnerability of Latent Layers in Adversarially Trained Models*. IJCAI 2019. [\[Paper\]](#).

- Puneet Mangla*, **Nupur Kumari***, Mayank Singh*, Abhishek Sinha*, Balaji Krishnamurthy, Vineeth N Balasubramanian. *Charting the Right Manifold: Manifold Mixup for Few-shot Learning*. Spotlight at MetaLearn, NeurIPS Workshop 2019. [\[Paper\]](#).

(* equal contribution)

US Patents

- **Nupur Kumari**, Richard Zhange, Junyan Zhu, Elya Shechtman. Fine-tuning and controlling diffusion models. (US20240185588A1)
- **Nupur Kumari**, Piyush Gupta, Akash Rupela, Siddarth R, Balaji Krishnamurthy, Bishal Deb, Ankita Sarkar. Generating a high-dimensional network graph for data visualization utilizing landmark data points and modularity-based manifold tearing. (US11295491B2)
- Balaji Krishnamurthy, Piyush Gupta, **Nupur Kumari**, Akash Rupela. Facilitating machine learning and data analysis by computing user-session representation vectors. (US10726325B2)
- Mayank Singh, **Nupur Kumari**, Dhruv Khattar, Balaji Krishnamurthy, Abhishek Sinha. Generating trained neural networks with increased robustness against adversarial attacks. **US11829880B2**

Press and Invited Talks:

- AIM Workshop, ICCV 2025, [\[Link\]](#)
- HiGen Workshop, ICCV 2025, [\[Link\]](#)
- Presented SynCD at Great Lakes Graphics Workshop. [\[Link\]](#)
- Presented Custom Diffusion at The AI Talks. [\[Link\]](#)
- Concept Ablation featured in CMU News. [\[Link\]](#)
- Custom Diffusion contributed to the AdobeFirefly custom model feature. [\[Link\]](#)

Award and Honors

- WiGRAPH Rising Star in Computer Graphics 2025. [\[Link\]](#)
- Top reviewer NeurIPS 2024.
- IIT Delhi Semester Merit Award, 2016.
- Top-30 in KVS-RMO and **INMO (Indian National Mathematics Olympiad)** 2012 participation award.

Academic Service

- Reviewer: CVPR, SIGGRAPH, ICCV, NeurIPS, ECCV, TPAMI, ICLR, IJCV, WACV.
- Organizer: Graphics Seminar at CMU. 2021-2026
- Teaching Assistantship:
 - Geometry-based methods in Vision Fall 2023, CMU
 - Learning for 3D Spring 2023, CMU
 - Machine Learning Adobe, India.
 - Linear Algebra Spring 2016, IIT Delhi
 - Discrete Mathematics Fall 2016, IIT Delhi
 - Data Mining Spring 2017, IIT Delhi
- Electrical coordinator, Robotics Club, IIT Delhi 2014-2015

Relevant Courses

Graduate:

Intro to Machine Learning
 Computer Vision
 Computational Photography
 Learning for 3D
 Geometry-based methods in Vision
 Physics Based Rendering

Undergraduate:

Digital Image Processing
 Discrete Mathematics
 Principles of Artificial Intelligence
 Natural Language Processing
 Computational Perception and Cognition